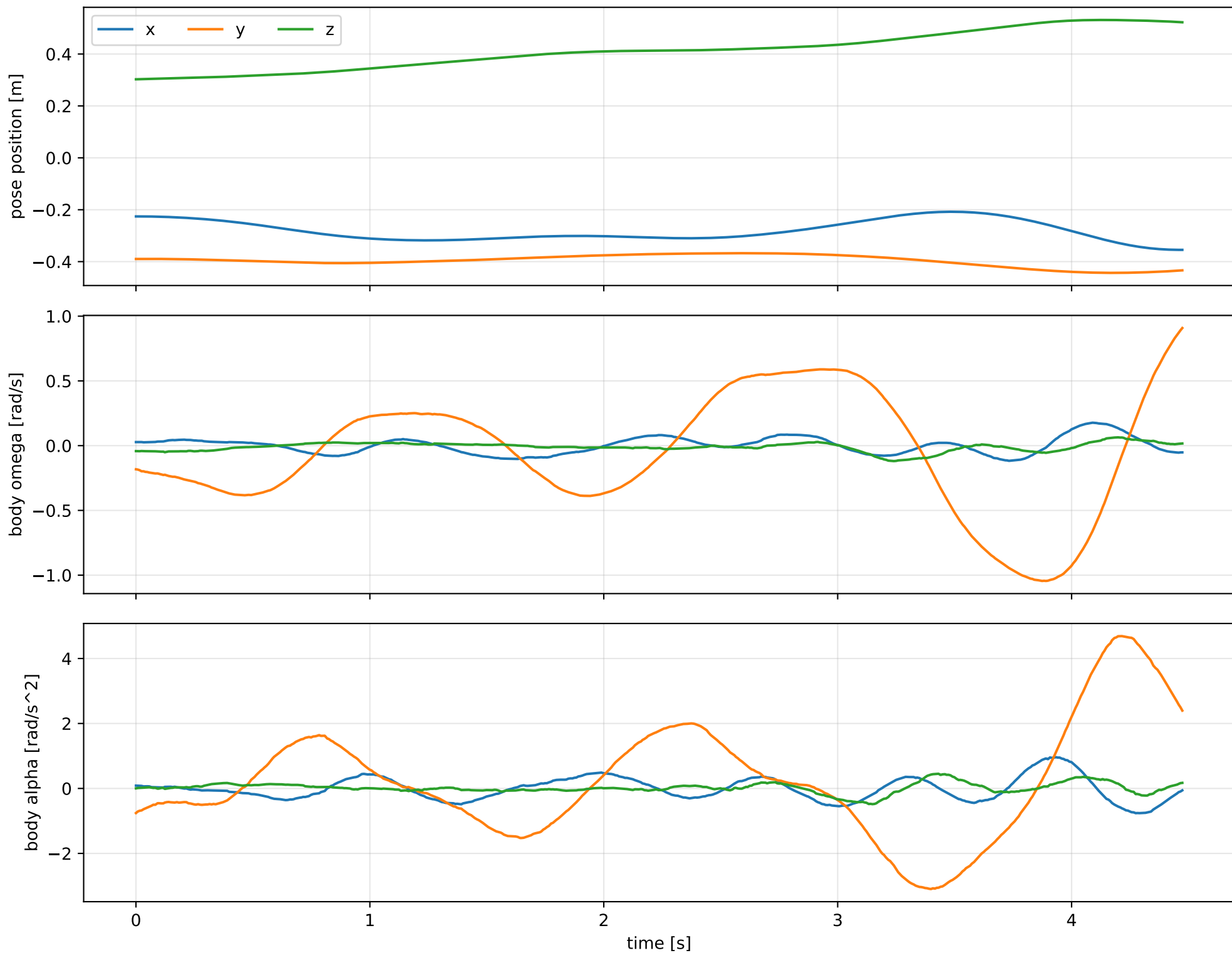
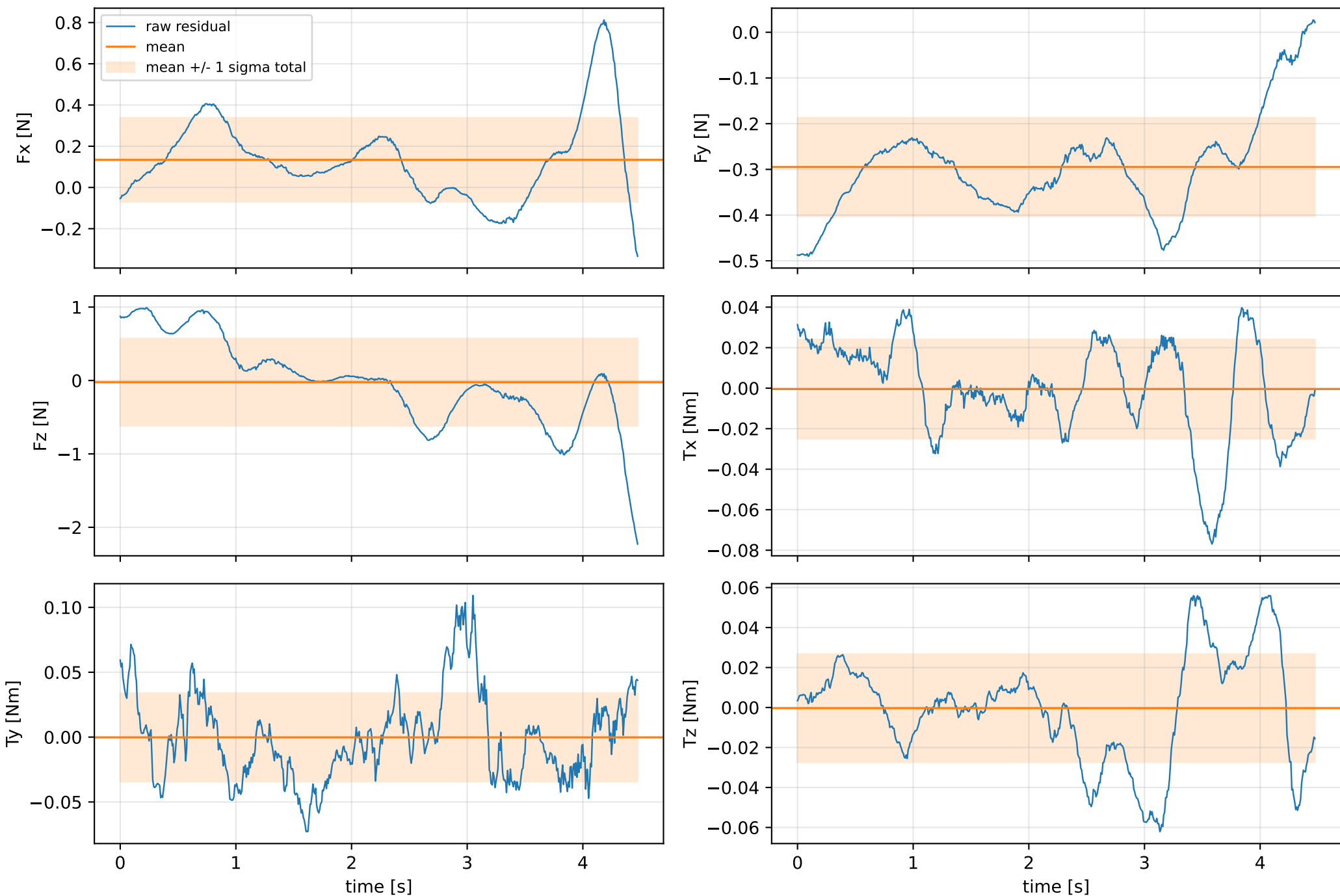


inertia_over_mass_xx_nominal: SG trajectory and derived motion

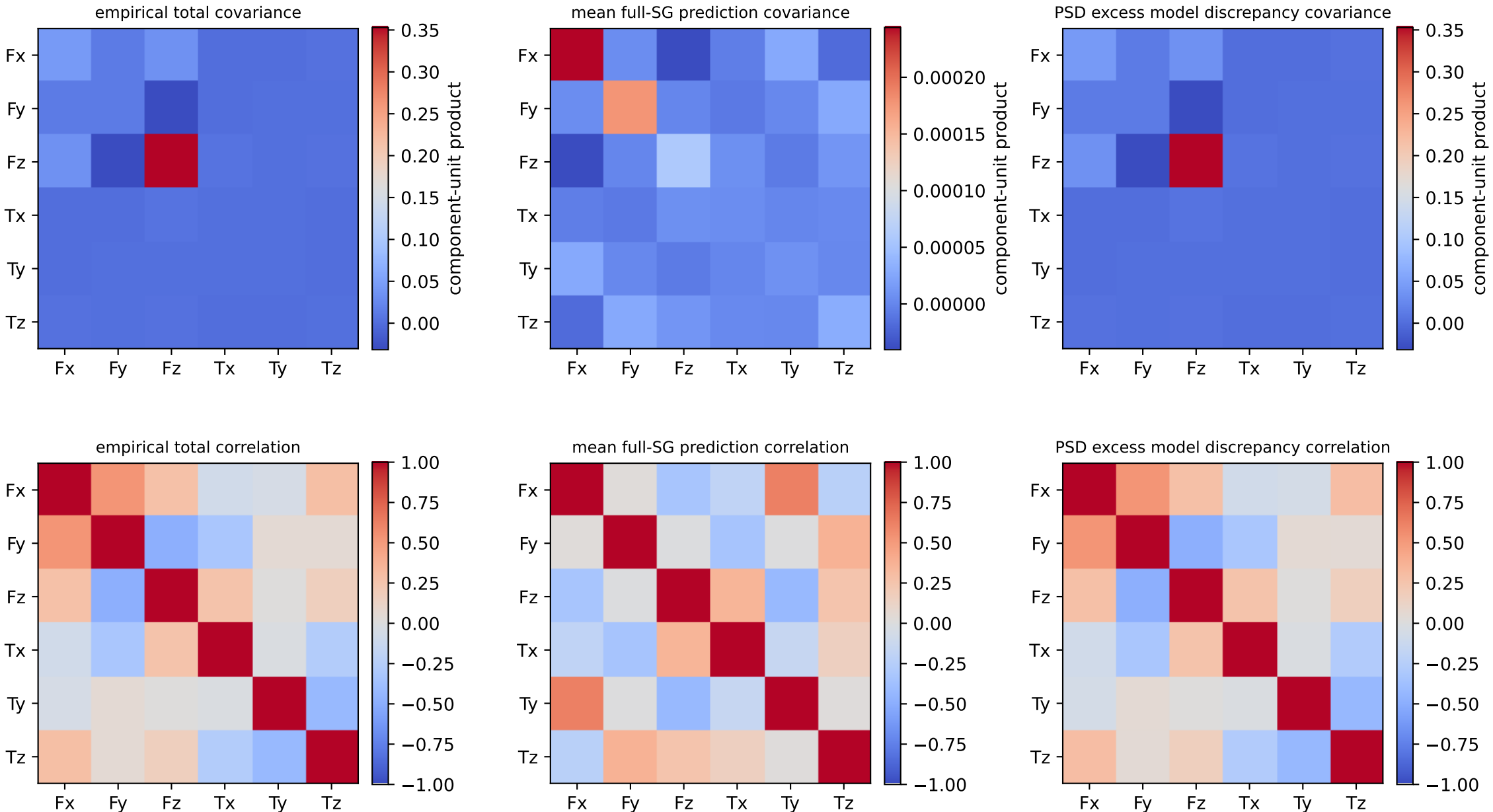


inertia_over_mass_xx_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



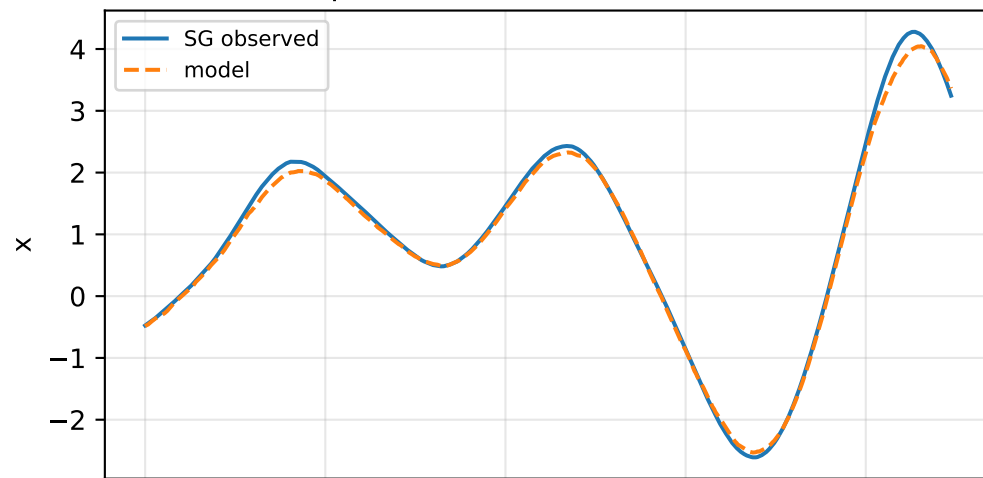
vector RMS about zero: force 0.71449 N, torque 0.04985 Nm; component std about mean: force [0.203 0.1076 0.5942], torque [0.0245 0.0341 0.027]

inertia_over_mass_xx_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [2.913e-04 6.427e-04 1.267e-03 2.868e-03 4.362e-02 3.591e-01]

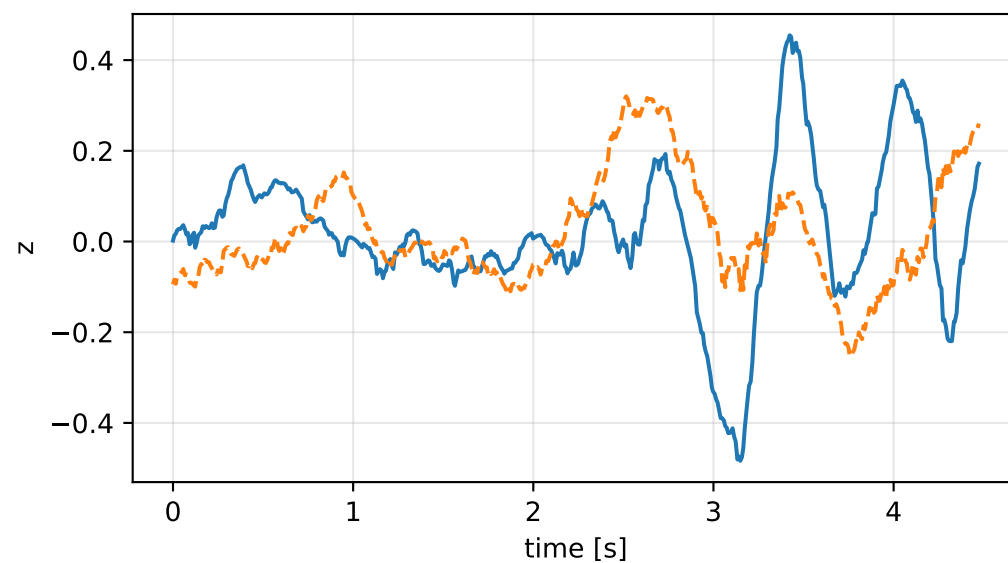
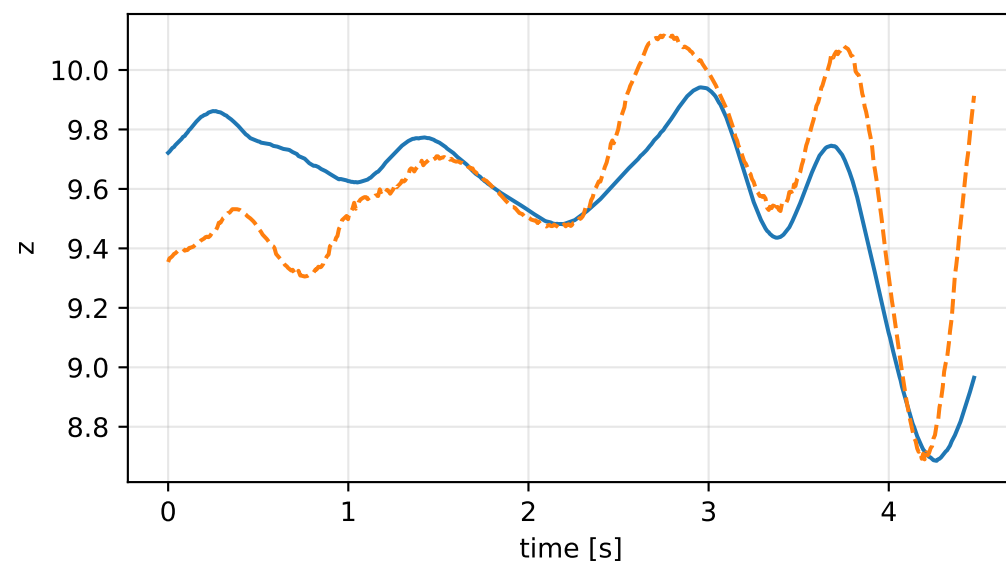
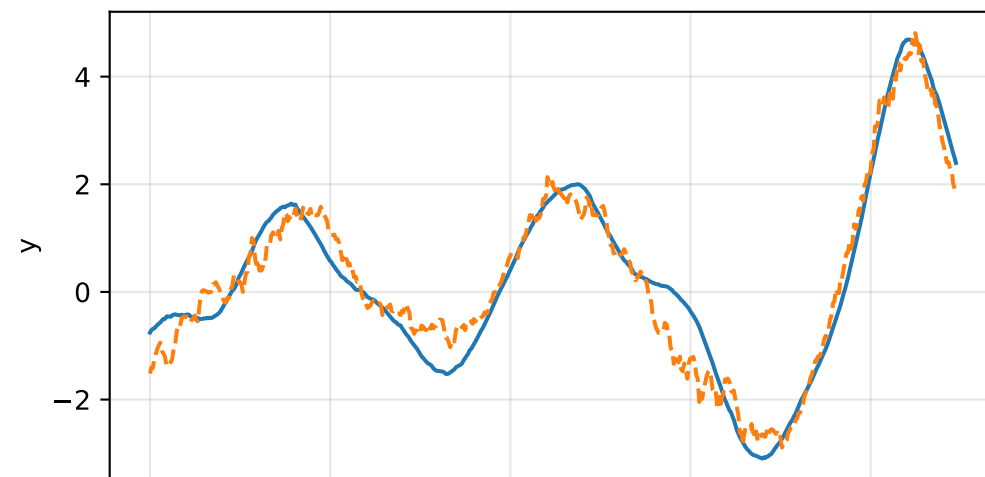
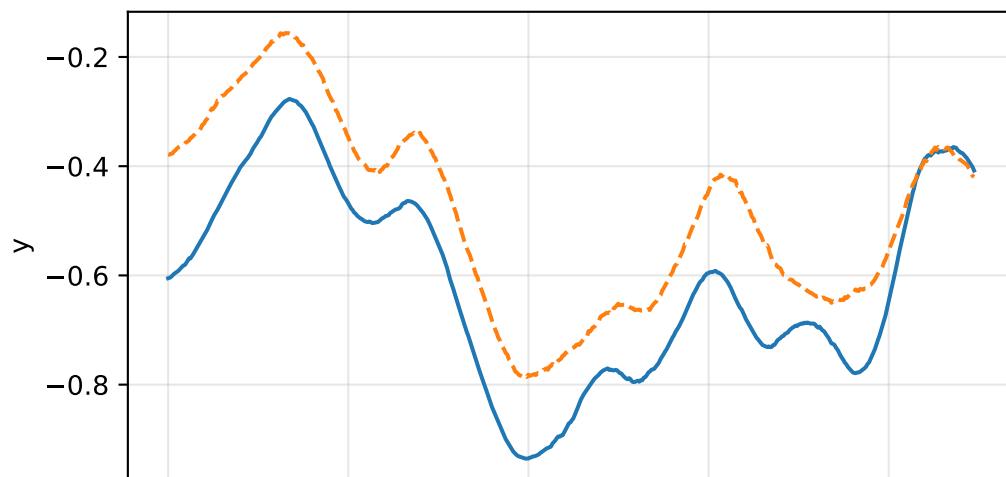
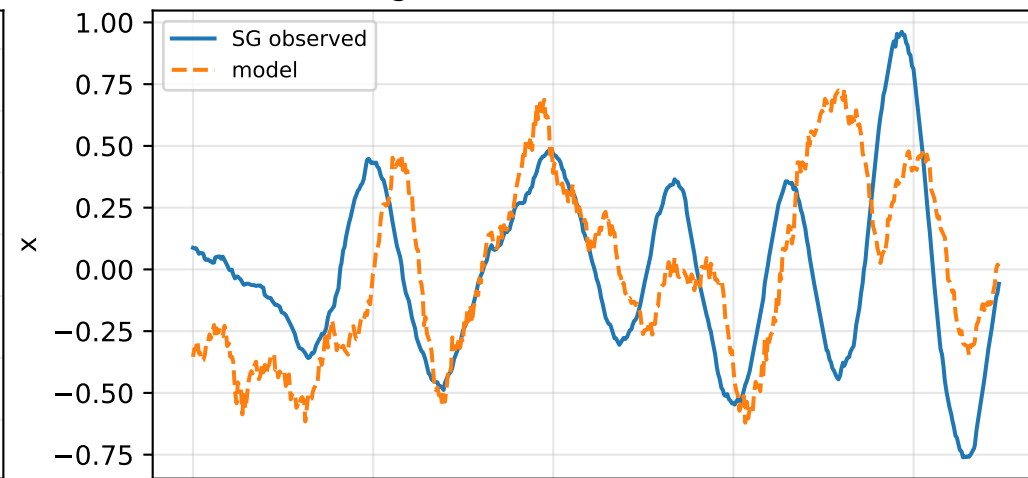


inertia_over_mass_xx_nominal: acceleration residual objective

specific acceleration [m/s^2]

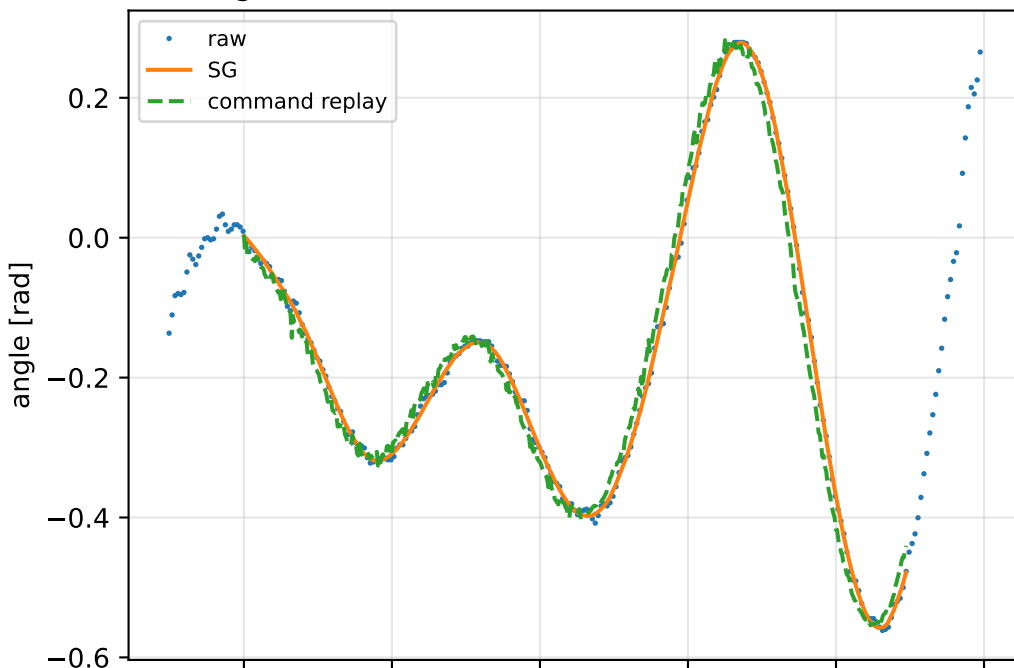


angular acceleration [rad/s^2]

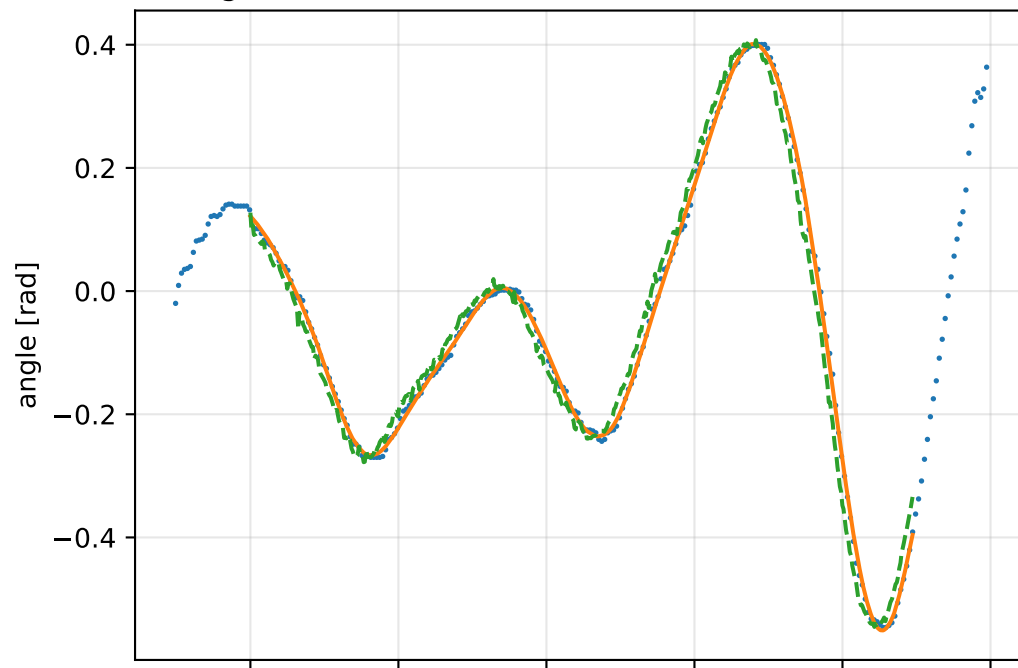


inertia_over_mass_xx_nominal: gimbal measurement audit (objective=measured_sg)

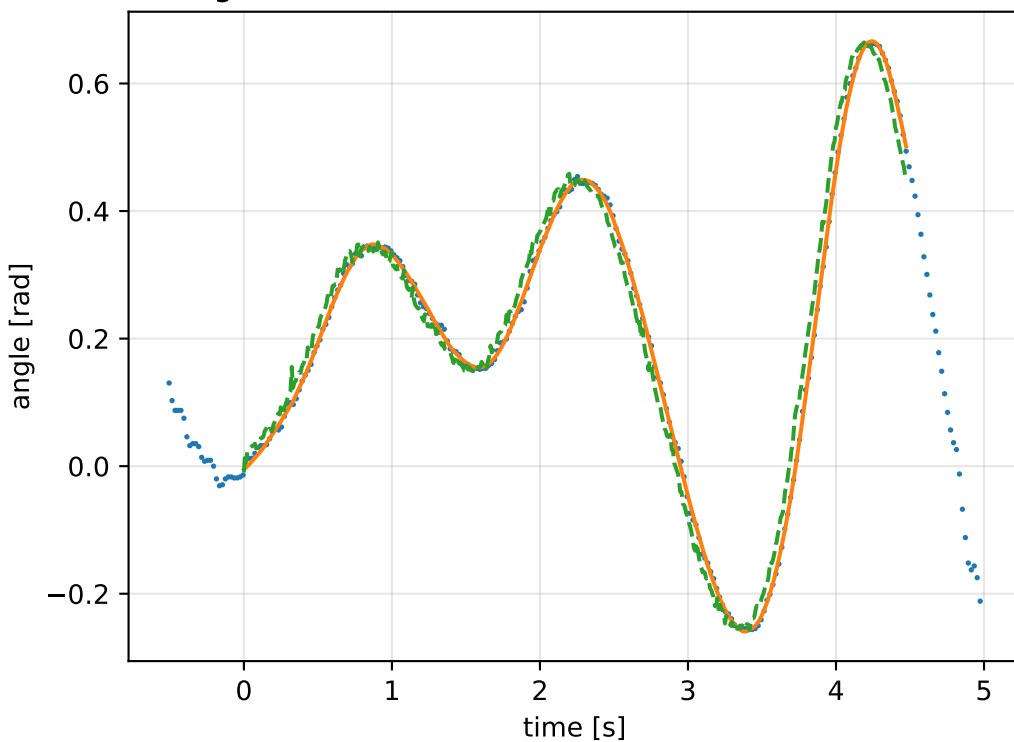
gimbal 1: RMSE=0.027, max=0.07529 rad



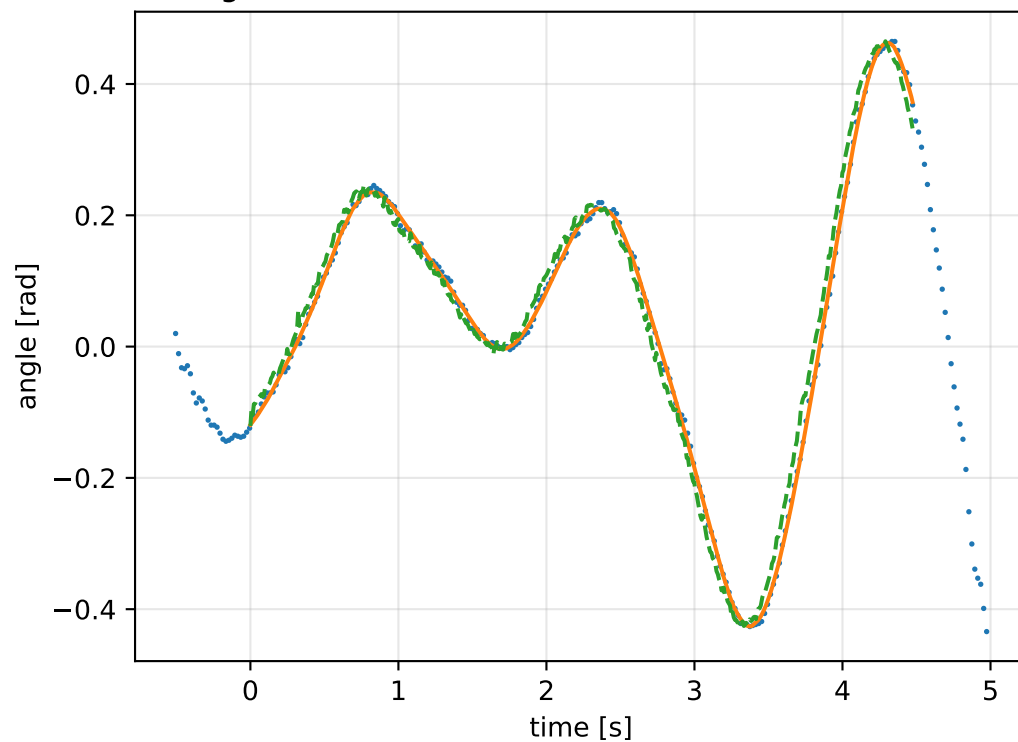
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

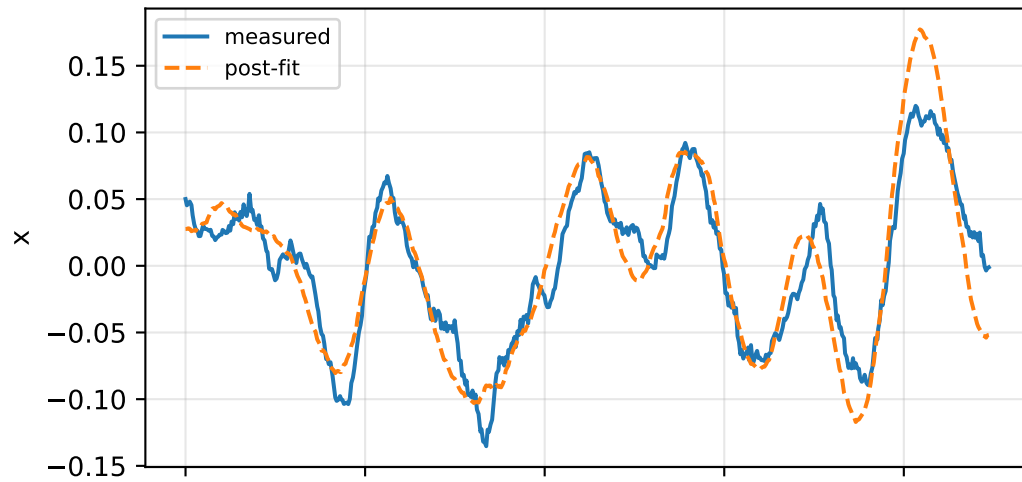


gimbal 4: RMSE=0.02598, max=0.07261 rad

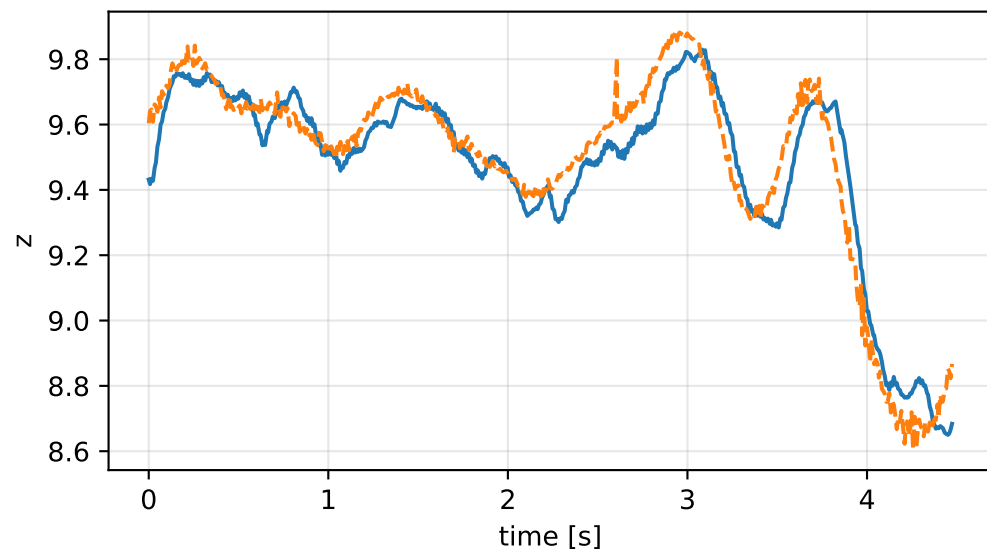
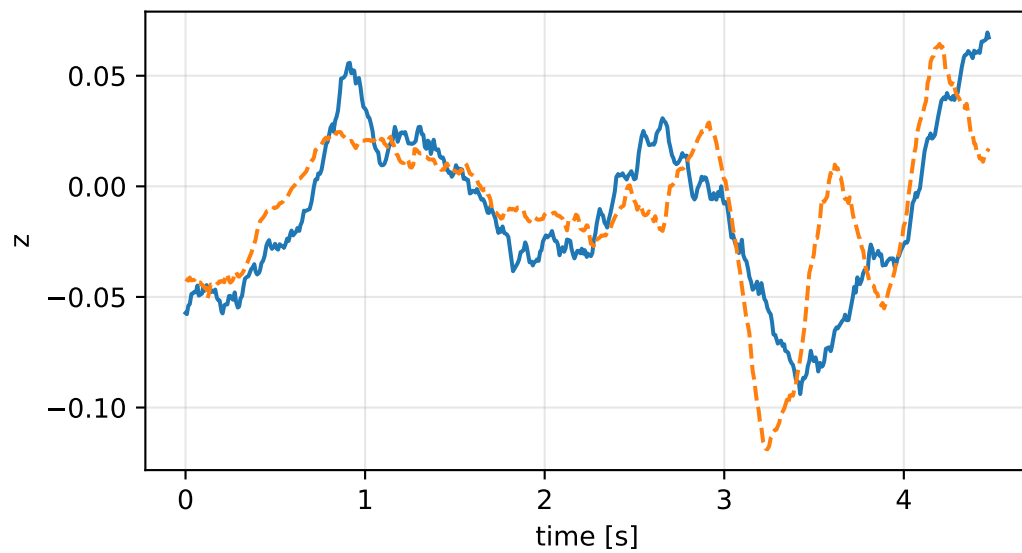
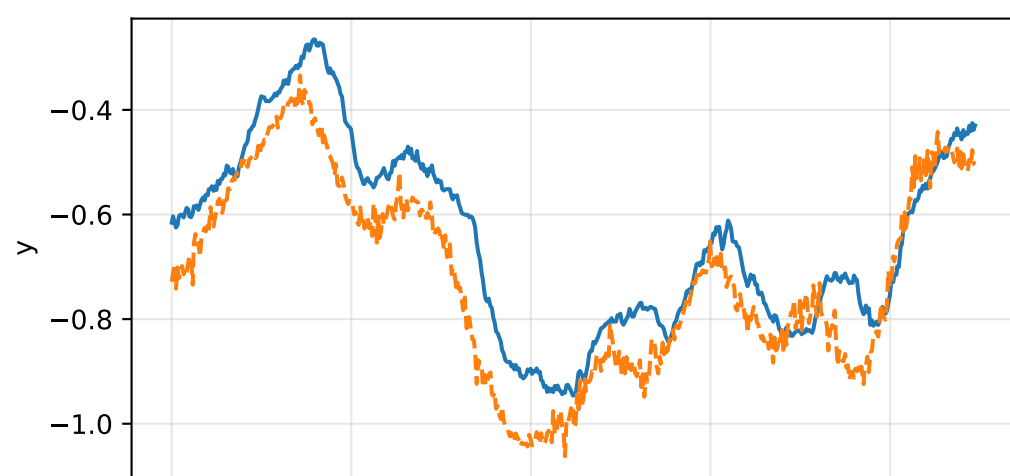
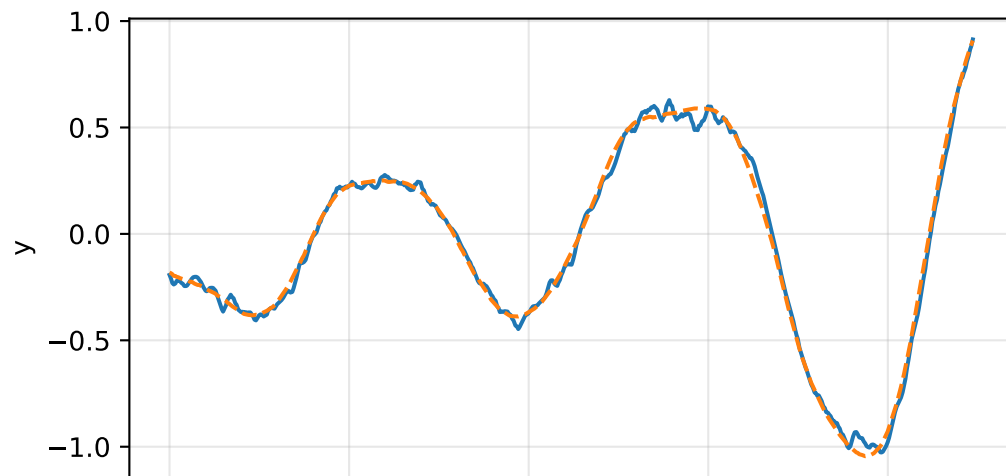
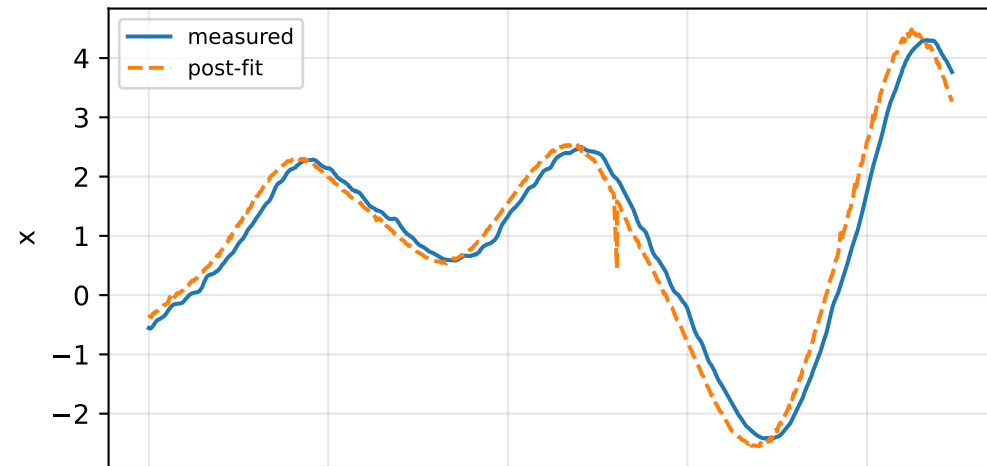


inertia_over_mass_xx_nominal: IMU comparison (diagnostic only)

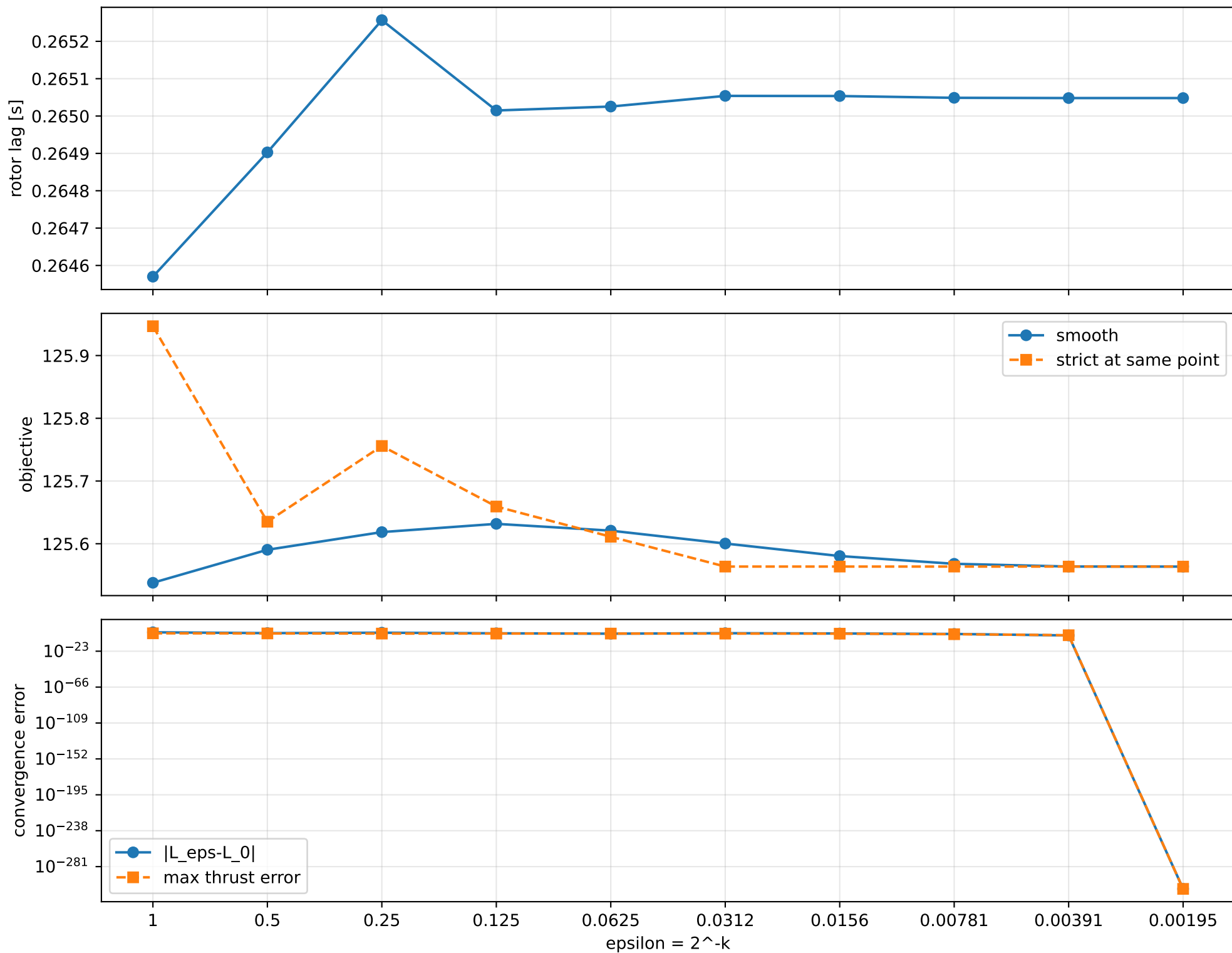
gyro [rad/s]



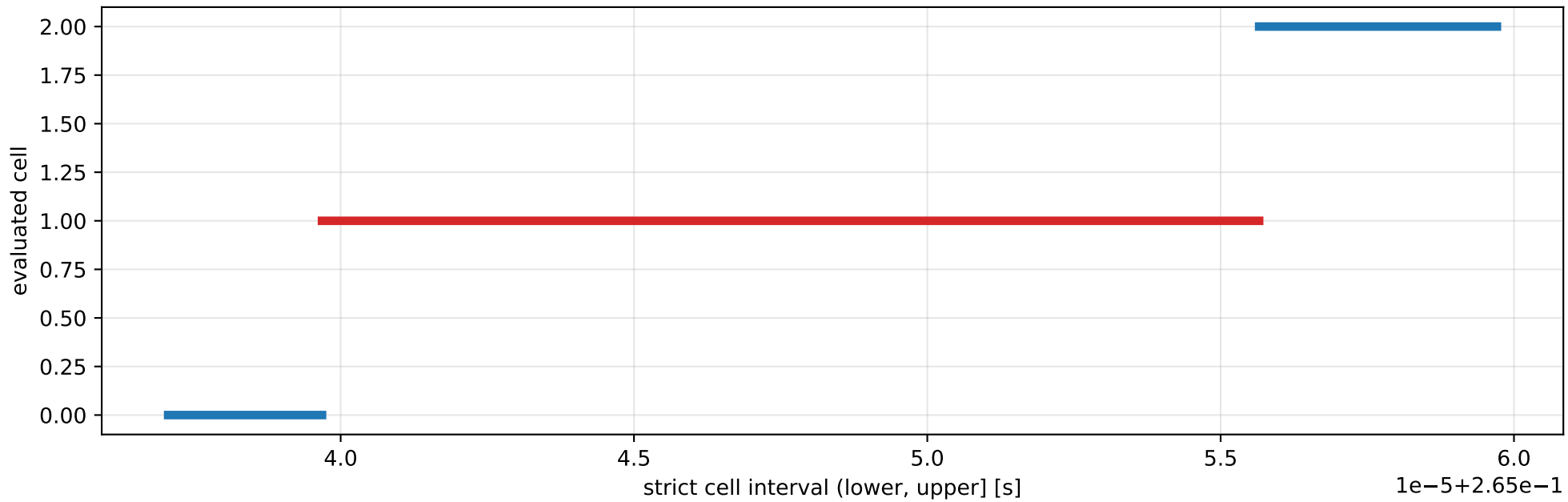
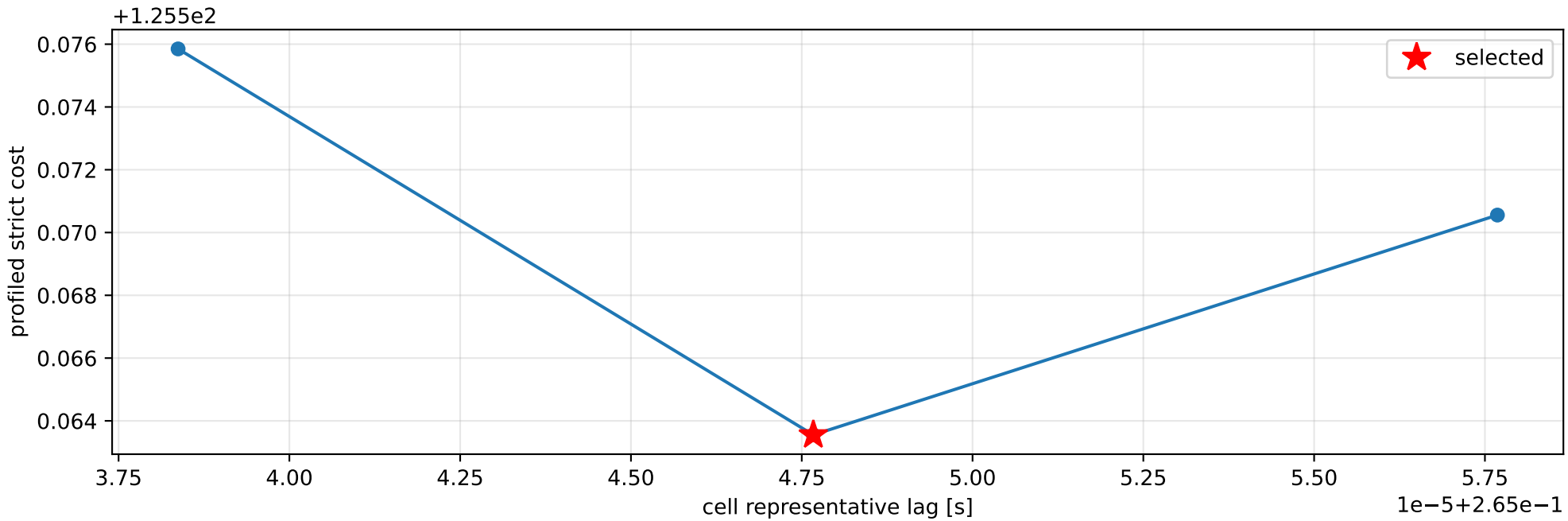
specific force [m/s^2]



inertia_over_mass_xx_nominal: smooth-to-strict continuation



inertia_over_mass_xx_nominal: exact strict-ZOH lag cells



selected (0.265039682, 0.265055656] s; final smooth 0.265048148 s; data boundary=False

inertia_over_mass_xx_nominal: parameter estimate

Case: inertia_over_mass_xx_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.90278863

inertia [kg m²]:

```
[[ 0.13551705  0.00970428 -0.01989102]
 [ 0.00970428  0.15931462 -0.01527636]
 [-0.01989102 -0.01527636  0.28551233]]
```

CoG body [m]: [-0.01993957 0.00433554 0.01214685]

force effectiveness: [1.842075 1.65435446 1.63395686 1.93708156]

scale-free J/m [m²]:

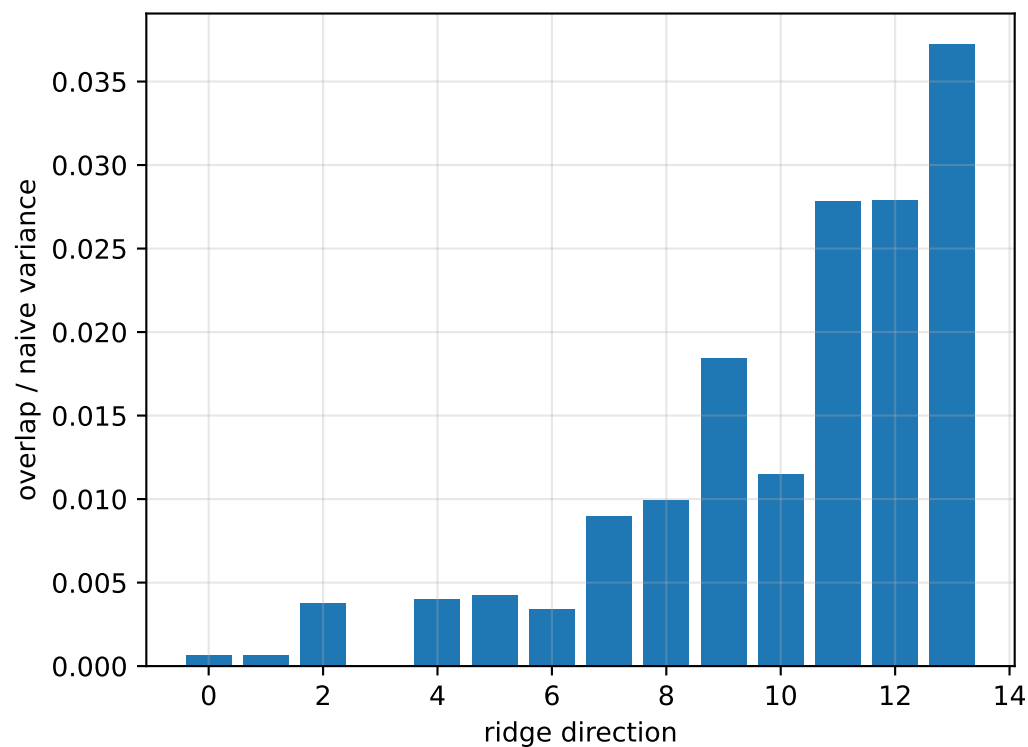
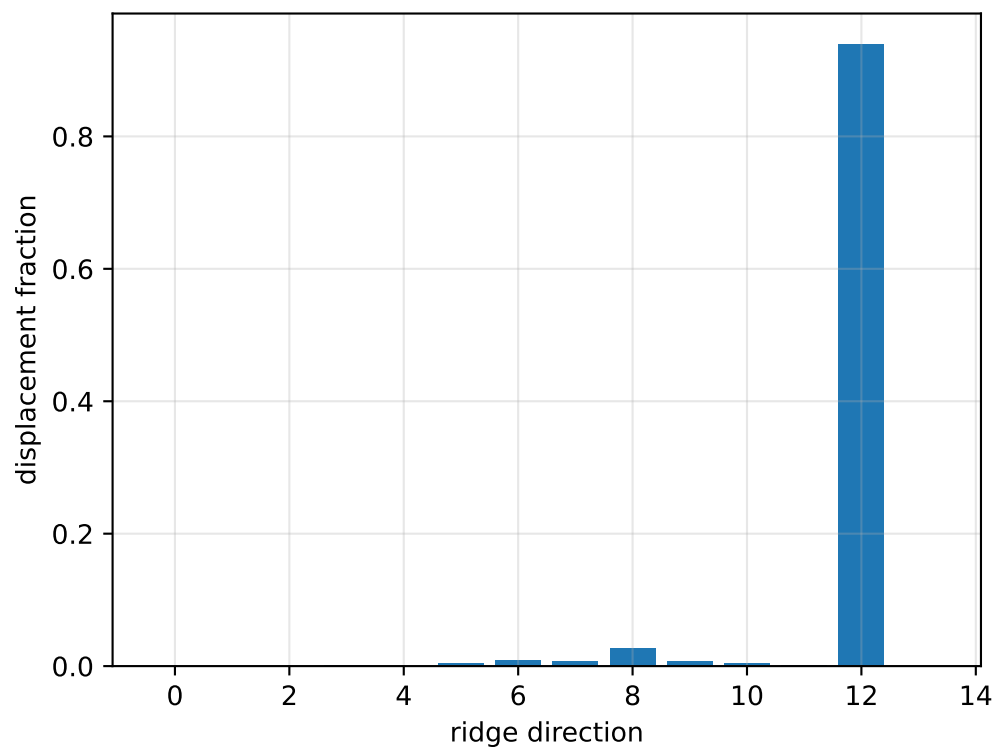
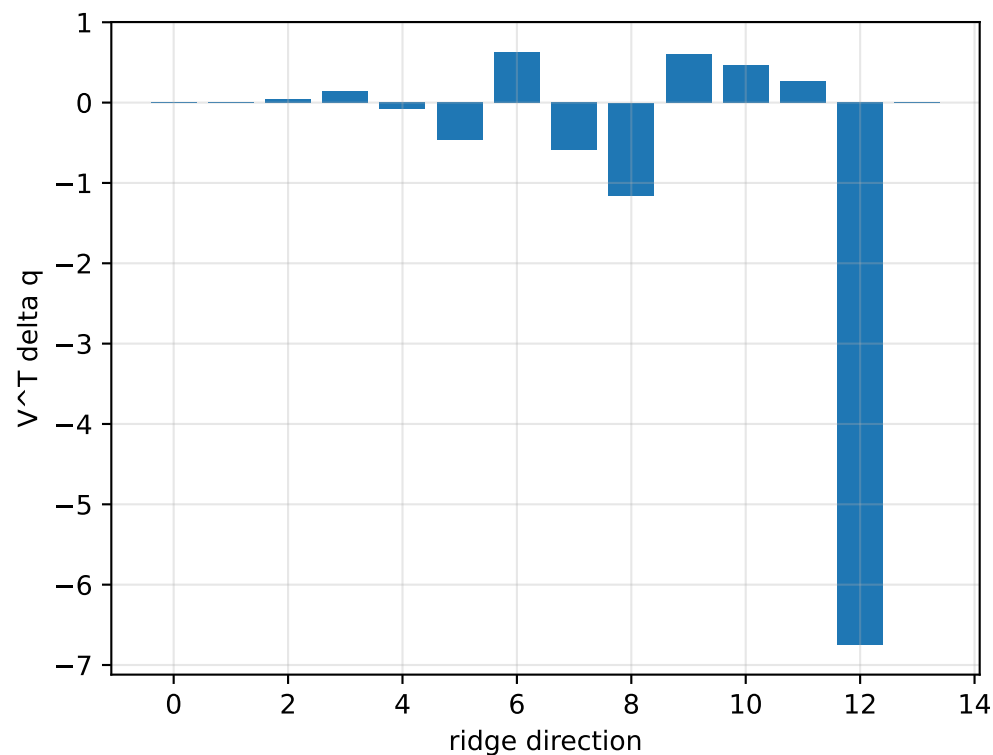
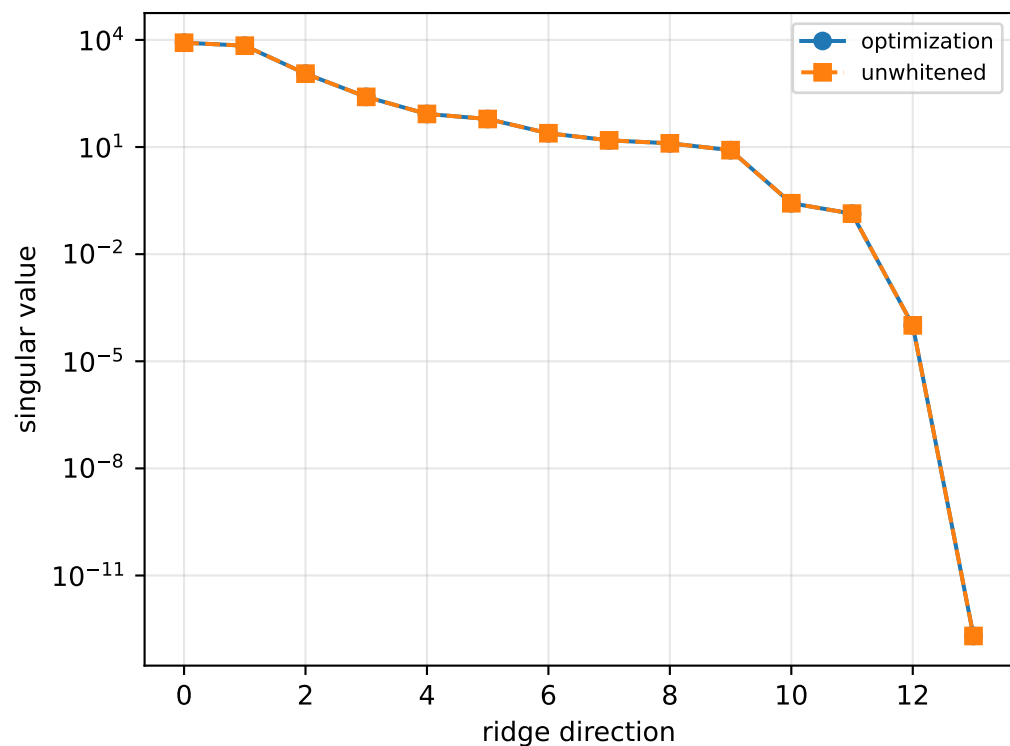
```
[[ 0.02764081  0.00197934 -0.00405708]
 [ 0.00197934  0.03249469 -0.00311585]
 [-0.00405708 -0.00311585  0.05823468]]
```

scale-free f/m: [0.37571985 0.33743132 0.33327092 0.39509791]

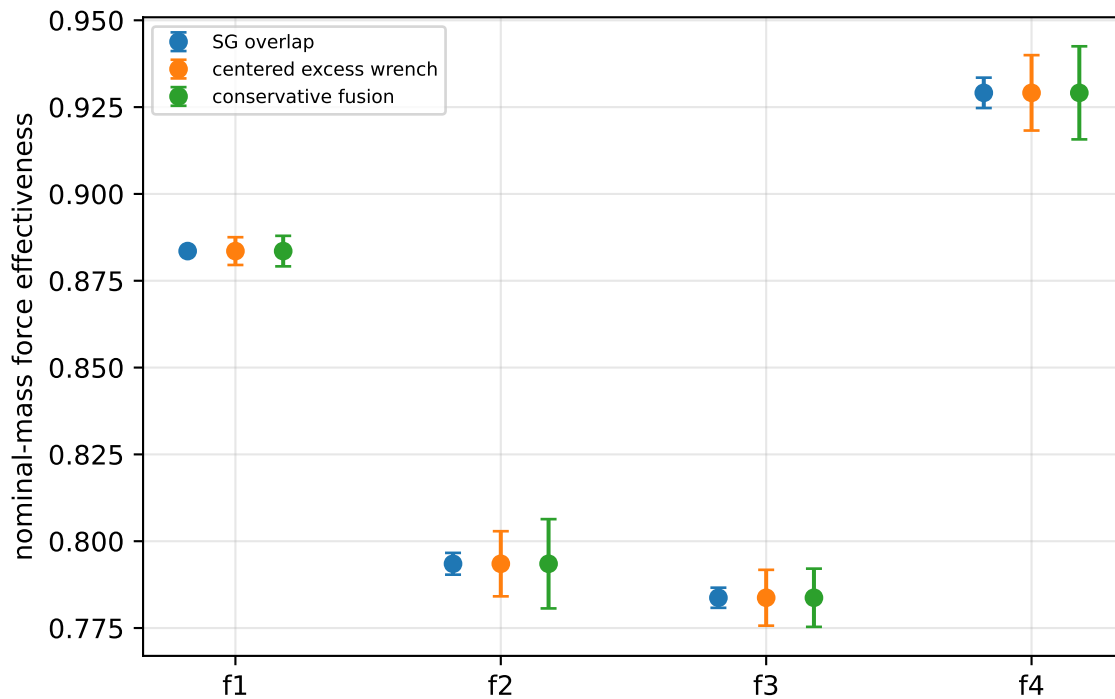
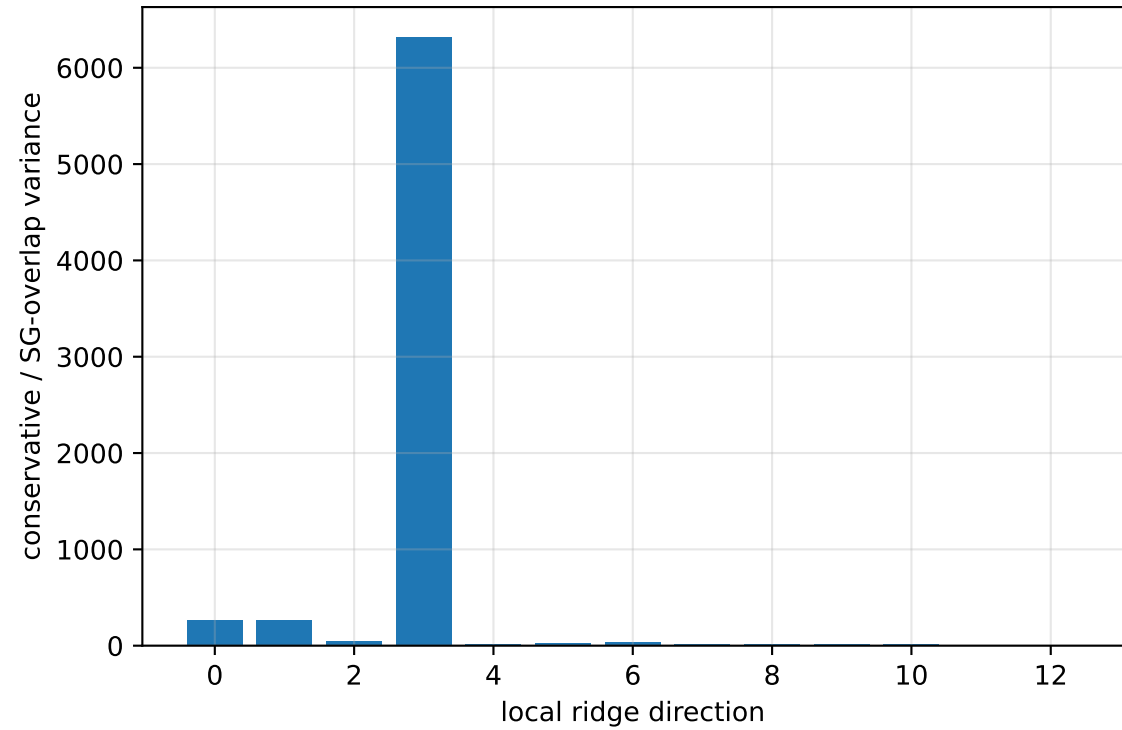
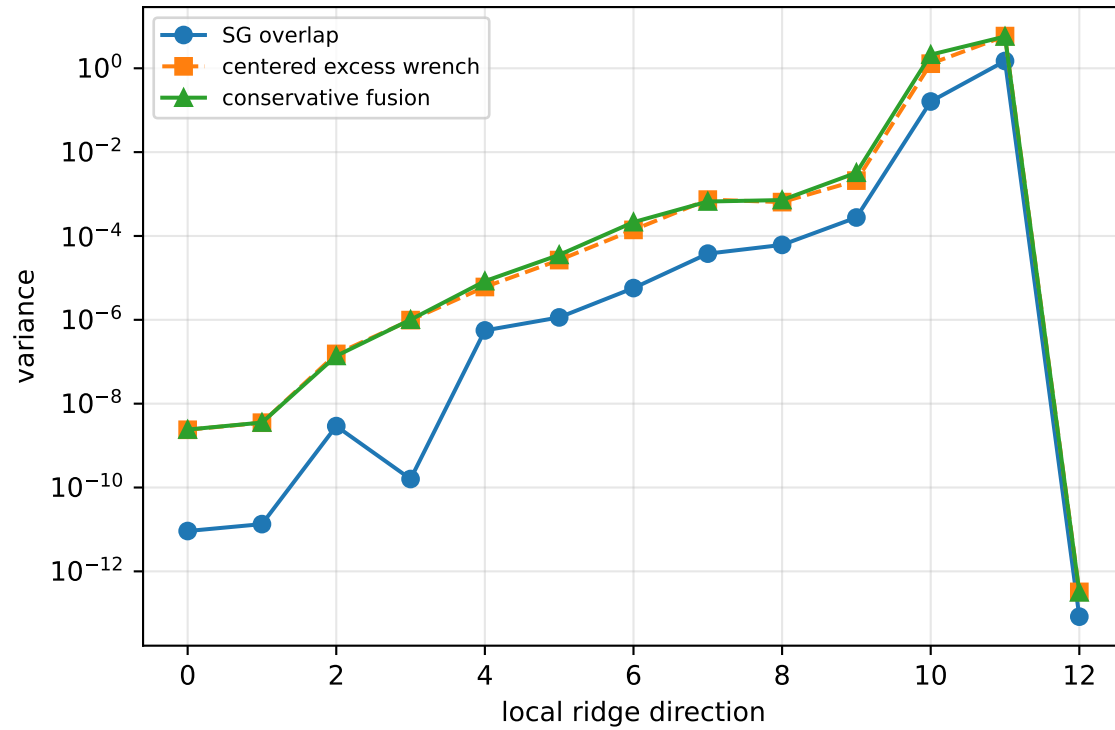
rotor lag cell: (0.265039682, 0.265055656] s

representative: 0.265047669 s

inertia_over_mass_xx_nominal: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=2.3\text{e-}13$)



inertia_over_mass_xx_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 249.4$
wrench-to-acceleration recovery max error = $2.89\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

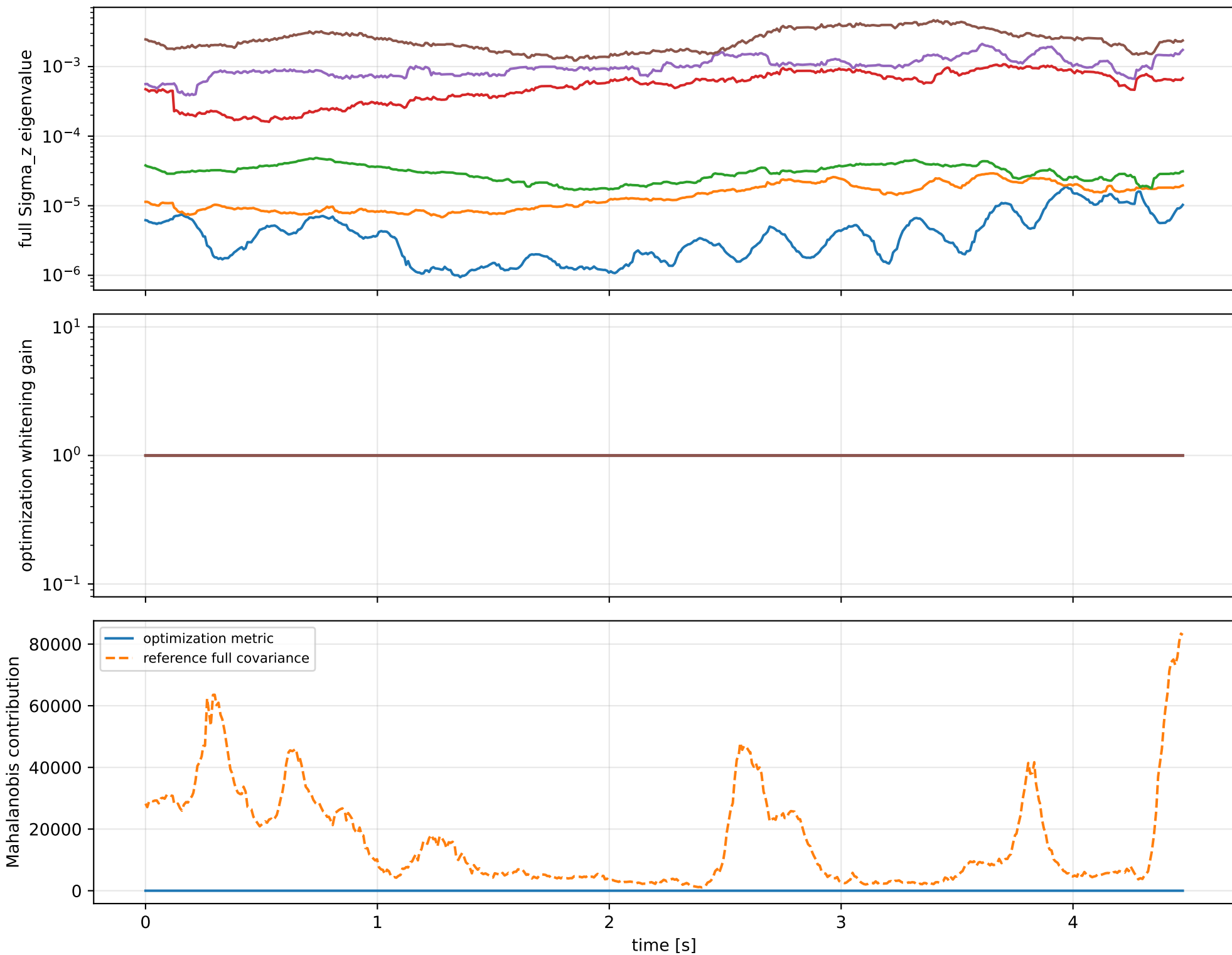
index 11: variance=5.723, ratio=3.831
 $\log_{\text{mass}} = -0.0196$; second_moment_diag_1=-0.664
second_moment_diag_2=+0.41; second_moment_diag_3=+0.352
second_moment_offdiag_12=-0.0715; second_moment_offdiag_13=+0.156
second_moment_offdiag_23=-0.485; cog_x=+3.57e-05
cog_y=-0.000111; cog_z=+2.66e-05
log_force_eff_1=-0.0195; log_force_eff_2=-0.0186
log_force_eff_3=-0.0203; log_force_eff_4=-0.02

index 10: variance=2.111, ratio=13.09
 $\log_{\text{mass}} = -0.0395$; second_moment_diag_1=+0.0372
second_moment_diag_2=-0.493; second_moment_diag_3=+0.654
second_moment_offdiag_12=-0.293; second_moment_offdiag_13=-0.475
second_moment_offdiag_23=-0.095; cog_x=-0.000293
cog_y=+0.000709; cog_z=-1.04e-05
log_force_eff_1=-0.0403; log_force_eff_2=-0.0455
log_force_eff_3=-0.0365; log_force_eff_4=-0.0365

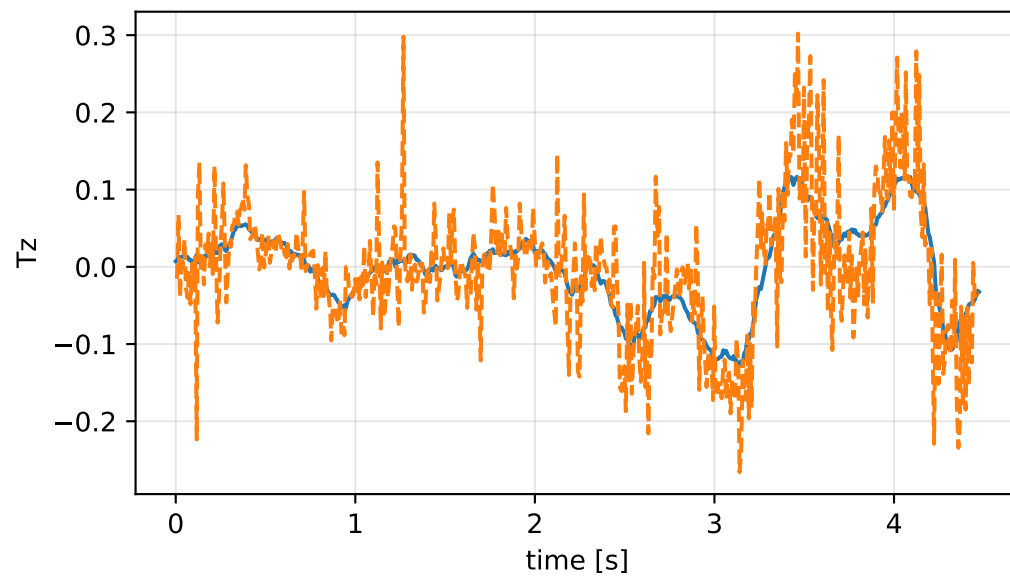
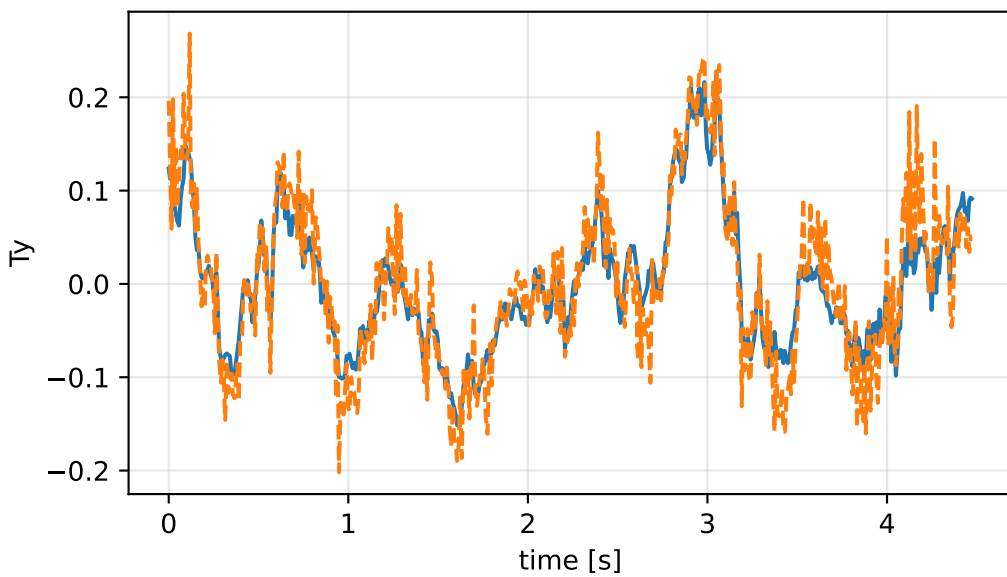
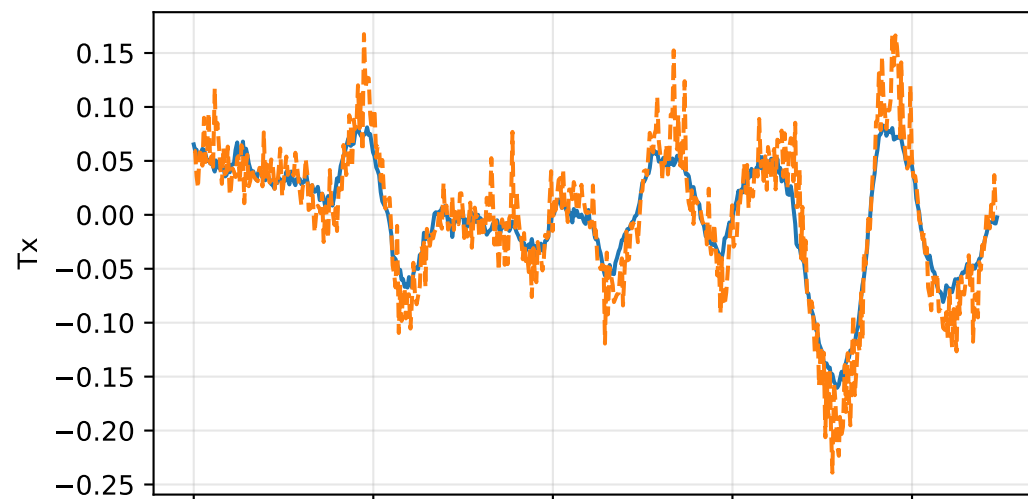
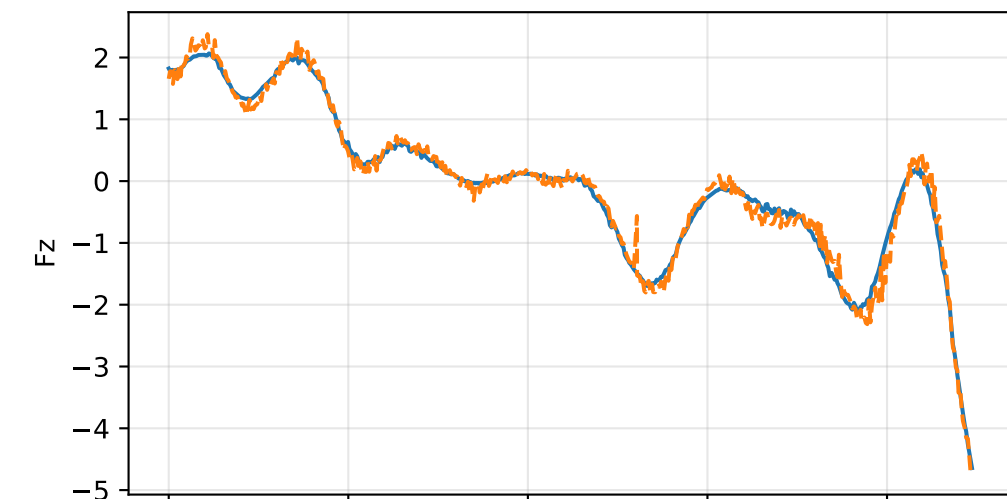
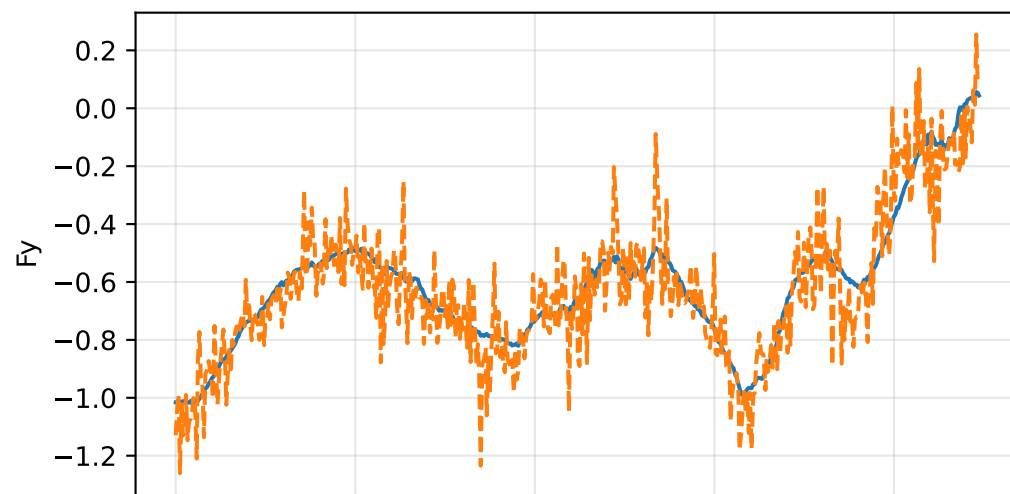
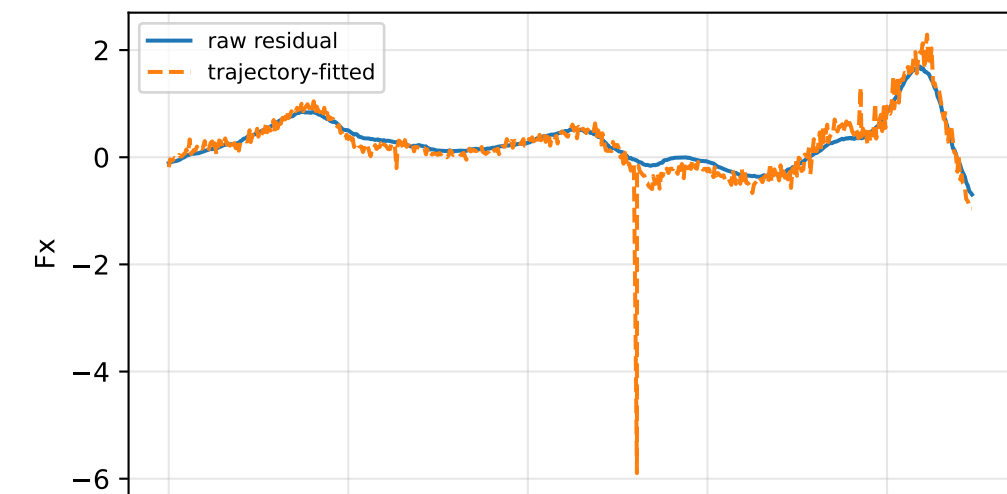
index 9: variance=0.003222, ratio=11.68
 $\log_{\text{mass}} = -0.158$; second_moment_diag_1=+0.0877
second_moment_diag_2=+0.586; second_moment_diag_3=+0.167
second_moment_offdiag_12=-0.319; second_moment_offdiag_13=-0.206
second_moment_offdiag_23=+0.512; cog_x=-0.0283
cog_y=+0.00608; cog_z=-0.00548
log_force_eff_1=-0.11; log_force_eff_2=-0.27
log_force_eff_3=-0.321; log_force_eff_4=+0.0178

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_xx_nominal: optimization vs reference covariance



inertia_over_mass_xx_nominal: wrench / closure (force max 1.31e-14, torque max 2.39e-16)



inertia_over_mass_xx_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_xx_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: d50f14634ccbffa93995c113d524958bbedaa410a60c8ea864b4c5cd041abdb4

data objective: 125.563552343

prior objective: 9.34693824571e-07

total objective: 125.563553278

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_xx_nominal (inertia_over_mass_m2)

components: xx

target: [0.02764081]

std: [1.e-06]

final: [0.02764081]

error: [1.36725552e-09]

standardized residual: [0.00136726]

factor objective: 9.34693824571e-07